

DESCRIPTION

LTG devices from Lotus Microsystems are thermally conductive yet electrically isolated, silicon-based thermal jumpers.

They are designed to guide heat away from hot electronic components, toward heat sinks or cooler areas (such as ground planes), without establishing an electrical connection.

LTG devices significantly enhance thermal conductivity, particularly in scenarios with limited or no direct access to a ground plane or heat sink, such as high-side switches in a half-bridge configuration.

Silicon, as an alternative to traditional ceramic materials, offers a cost-effective substrate with high thermal conductivity and excellent thermo-mechanical properties and reliable processing. The incorporation of LTG devices both improves circuit reliability and reduces the overall cost of a thermal management system.

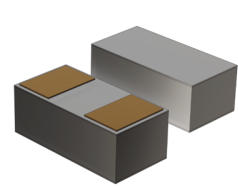
They are currently available in standard EIA sizes (0201, 0402, 0603, 0805, 1206) and custom sizes can be made available on request.

FEATURES

- High thermal conductivity
- High insulation resistance
- Low parasitic
- High precision package sizing
- Low coefficient of thermal expansion (CTE)
- RoHS compliant
- Thermal Conductivity up to 140 W/(m K)

APPLICATIONS

- High-power LEDs
- Pin and laser diodes
- Power converters and amplifiers
- xPU and FPGA boards
- Semiconductor packages
- Optical Transceivers



Thermal Performance

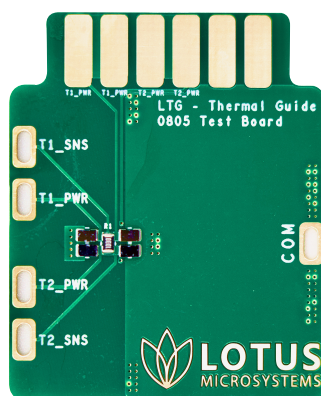


Figure 1: Top view of evaluation board

The LTGs are characterized using a thermal camera. The evaluation board is a 35 mm × 35 mm × 1.6 mm, 2-layer board with a copper thickness of 35 μm. The target cooling device is a resistor connected to a power supply and is stressed such that the surface temperature stabilizes at approximately 140 °C with no thermal aid, creating the reference case when populating the LTG devices. More details about the evaluation board can be found in the LTG Evaluation Board Manual.

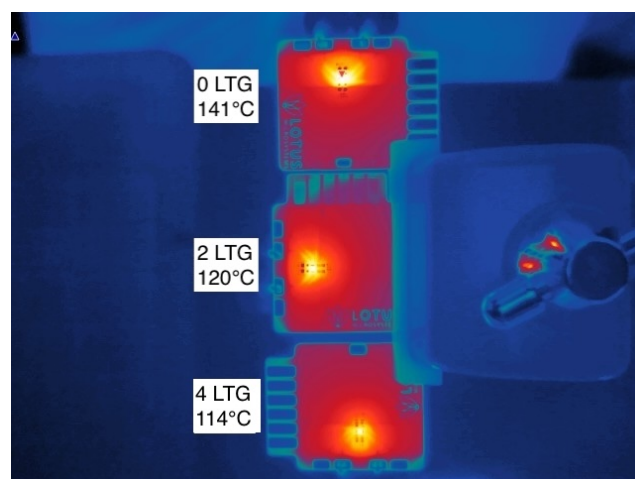


Figure 2: Comparing 0, 2 and 4 LTG0603 thermal performance

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1 Absolute Maximum Rating

All pins to GND¹ 0 V to 150 V
Operating Temperature Range -65 °C to 150 °C
Junction Temperature Range -65 °C to 150 °C
Maximum Reflow Temperature 260 °C

2 Material Composition

Substrate Material	Silicon (150 W/(m K))
Termination Material	Copper with solder balls
Solder Material	SnAg (1.8 %)

Table 1

3 Typical Characteristics

All values are measured or simulated at $T_A = 25\text{ °C}$ unless otherwise specified

Case Size	Length (mm)	Width (mm)	Height (mm)	Terminal pins (mm)	Thermal Resistance w/ solder °C/W	Thermal Resistance w/o solder °C/W
0201	$0.6 \pm 3\%$	$0.3 \pm 3\%$	$0.3 \pm 3\%$	0.20×0.29	N/A	75.0^2
0402	$1.0 \pm 3\%$	$0.5 \pm 3\%$	$0.8 \pm 3\%$	0.24×0.43	42.2	29.7
0603	$1.6 \pm 3\%$	$0.8 \pm 3\%$	$0.8 \pm 3\%$	0.40×0.70	19.0	15.0
0805	$2.0 \pm 3\%$	$1.2 \pm 3\%$	$0.8 \pm 3\%$	0.50×1.10	11.3	9.4
1206	$3.2 \pm 3\%$	$1.6 \pm 3\%$	$0.8 \pm 3\%$	0.83×1.56	N/A	9.0^2

Table 2: Physical characteristics

4 Electrical Characteristics

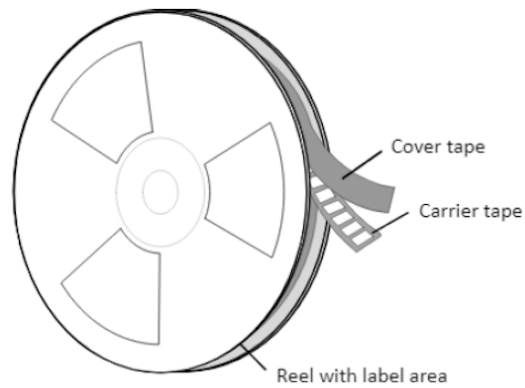
Case Size	Capacitance (pF)	Isolation Resistance (GΩ)
0201	0.55	1.0
0402	1.4	2.1
0603	3.9	0.7
0805	5.6	0.5
1206	12.0	0.1

Table 3: Electrical characteristics

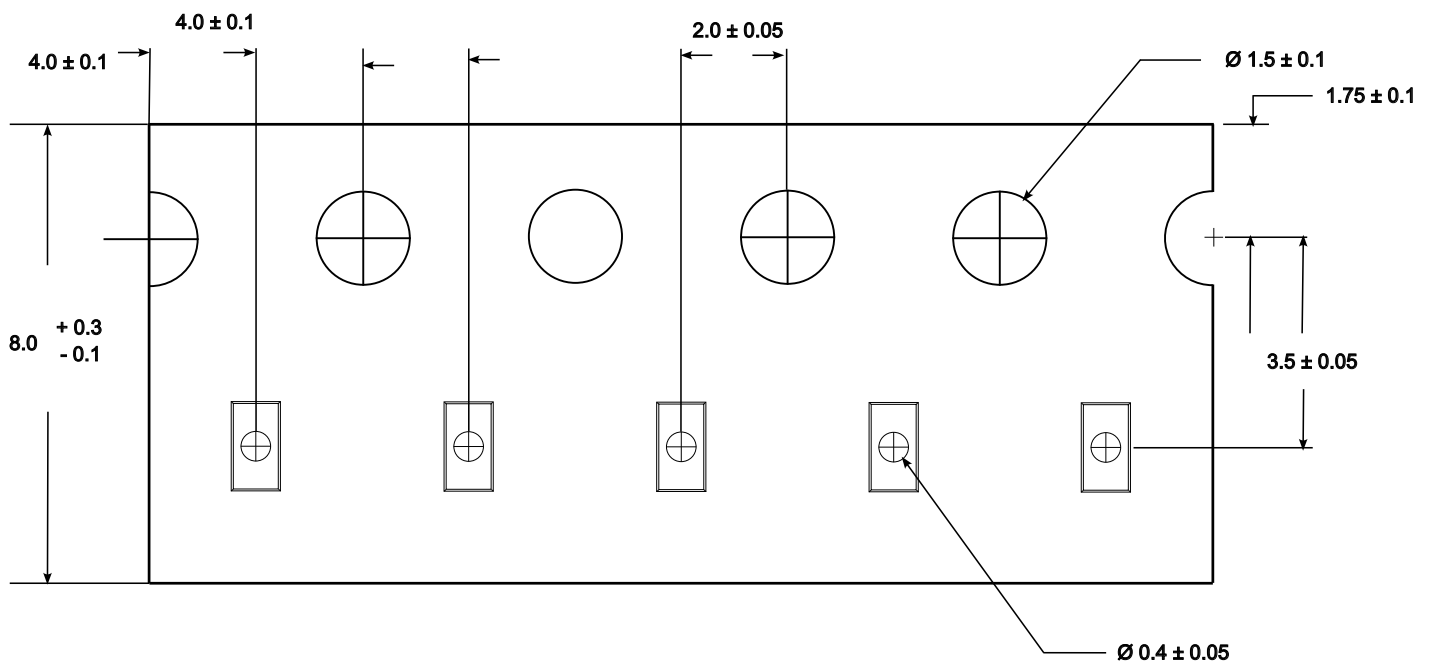
¹Exposing the device to conditions beyond specified in section 1 may affect the reliability of the device or cause permanent damage.

²Based on simulations.

5 Tape and Reel Information

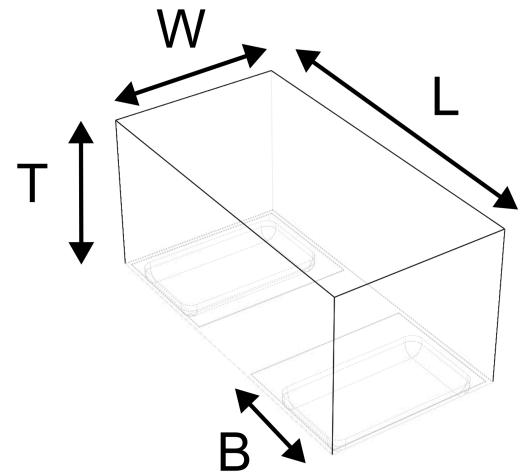
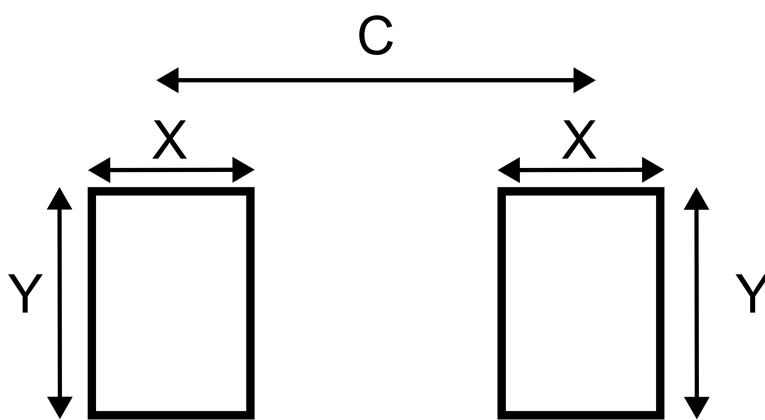


7 inch reel



All units are in mm

6 Package Information



Case size	L	W	T	B
0201	0.6	0.3	0.3	0.20
0402	1.0	0.5	0.8	0.24
0603	1.6	0.8	0.8	0.43
0805	2.0	1.2	0.8	0.50
1206	3.2	1.6	0.8	0.83

Table 4: Standard package sizes. All units are displayed in mm

Recommended footprint		C	X	Y
0201	Least	0.65	0.30	0.25
	Nominal	0.70	0.40	0.35
	Most	0.75	0.56	0.40
0402	Least	0.90	0.50	0.55
	Nominal	0.95	0.55	0.55
	Most	1.00	0.60	0.60
0603	Least	1.35	0.75	0.80
	Nominal	1.45	0.85	0.90
	Most	1.55	0.95	1.00
0805	Least	1.75	0.85	1.20
	Nominal	1.85	0.95	1.30
	Most	1.95	1.05	1.40
1206	Least	2.9	1.15	1.6
	Nominal	2.95	1.35	1.8
	Most	3.00	1.50	2.00

Table 5: Landing pattern (IPC7351). All units are displayed in mm

7 Part Number Information

L TG 0402 80 ST

Thermal Guide		Size		Thickness		Termination		Packaging	
		EIA	0402	80	0.80 mm	S	With Solder	T	Tape and Reel
		EIA	0603	TT	Custom (mm)	C	No Solder	C	Matrix Tray
		EIA	0805						
		Custom	LL * WW (mils)						

8 Changelog

Revision Number	Revision Date	Description	Sections Changed
1.0	2025-05-21	Initial release	-
1.1	2026-04-29	Added specifications for EIA size 0201 and 1206	3, 4, 6

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